TREEWALK NYC • STEAM LESSON PLAN

Purple Leaf Plum

Prunus cerasifera

Species background: Native to southeastern Europe and western Asia (the wild species is Prunus cerasifera); purple-leaved ornamental varieties were selectively bred and became popular street and yard trees in the 20th century. Like Crimson King maple, its purple color is present all season rather than just in fall -- a useful way to distinguish it from ordinary green-leaved cherry or plum trees at a glance.

This packet contains one full lesson for each grade band: K-2, 3-5, 6-8, and 9-12. Each includes a learning objective, standards connection, materials, teacher background, a step-by-step procedure, discussion questions, an assessment, and an extension activity.

GRADE BAND K-2

Purple Leaf Plum: Find purple leaf plum's purple football-shaped leaf.

Standards connection: NGSS K-LS1-1; National Core Arts Standards VA:Cr1 (K-2)

Materials: Coloring sheet, purple and pink crayons.

Teacher background

My leaves are purple all spring and summer, not just in fall! Right before my leaves come out, I might be covered in soft pink flowers.

Procedure

1. Engage: Ask students what shape a football looks like.
2. Explore: Compare this leaf's oval shape to a football.
3. Explain: Explain that this tree, like Crimson King maple, is purple all season, not just fall.
4. Art: Find a purple leaf shaped like a football and trace it.
5. Evaluate: Student can describe the leaf as purple and football-shaped.

Discussion questions

What season do you usually expect purple leaves, and why is this tree different?

Assessment

Correctly describes the leaf color and shape.

Extension

Look for pink flowers on this tree in spring, if in season.

GRADE BAND 3-5

Purple Leaf Plum: Compare purple leaf plum's small size to a large shade tree and explain why it fits small spaces.

Standards connection: NGSS 3-LS3-1; National Core Arts Standards VA:Cr2 (3-5)

Materials: Coloring sheet, paint mixing.

Teacher background

Sketch a small purple tree in springtime with pink blossoms just opening -- try mixing red and blue paint to make your own purple. It's a compact, small-statured tree.

Procedure

1. Engage: Ask why a small city tree pit might not fit every kind of tree.
2. Explore: Compare this tree's size to a large shade tree nearby.
3. Explain: Explain that its compact size made it popular for tight city spaces.
4. Art: Sketch a small purple tree in springtime with pink blossoms just opening -- try mixing red and blue paint to make your own purple.
5. Evaluate: Student explains in one sentence why a small tree might be chosen for a small space.

Discussion questions

What other small plants or trees have you seen fit into tight city spaces?

Assessment

Correct one-sentence explanation connecting small size to small spaces.

Extension

Find a tree pit on your walk that looks too small for a large shade tree.

GRADE BAND 6-8

Purple Leaf Plum: Explain convergent horticultural selection between unrelated purple-leaf cultivars.

Standards connection: NGSS MS-LS3-2; National Core Arts Standards VA:Cr2/Re7 (6-8)

Materials: Coloring sheet, paint materials.

Teacher background

Purple-leaved cultivars exist in both the maple family (Crimson King) and the rose family (this tree) -- unrelated plants bred independently for the same anthocyanin-rich look.

Procedure

1. Engage: Why might people independently breed purple-leaved varieties of two completely unrelated trees?
2. Explore: Research which plant family each tree belongs to (Sapindaceae for maple, Rosaceae for plum).
3. Explain: Discuss convergent horticultural selection as a human-driven parallel to natural convergent evolution.
4. Art: Paint a spring study of the tree in bloom, layering pink blossoms over a purple-leaf background -- research complementary colors and explain your color choice.
5. Evaluate: Write a paragraph explaining convergent horticultural selection using both trees as examples.

Discussion questions

Is human-driven selective breeding for a trait 'the same thing' as natural convergent evolution, or fundamentally different? Why?

Assessment

Paragraph correctly explains the parallel purple-leaf breeding in two unrelated families.

Extension

Research one other ornamental trait that's been independently bred into multiple unrelated tree species.

GRADE BAND 9-12

Purple Leaf Plum: Evaluate whether independently bred purple-leaf traits count as true convergent evolution.

Standards connection: NGSS HS-LS3-2; National Core Arts Standards Proficient-level Creating/Responding (9-12)

Materials: Coloring sheet, color-swatch materials.

Teacher background

Purple-leaf cultivars exist independently in the maple family (Sapindaceae) and the rose family (Rosaceae); research the specific anthocyanin genes involved in each case and evaluate whether this counts as true convergent evolution or simply parallel human selection pressure.

Procedure

1. Engage: Does deliberate human breeding for a trait count as 'evolution' in the biological sense, or does it require a different term?
2. Explore: Research the anthocyanin genetics involved in ornamental purple-leaf breeding, if documented.
3. Explain: Argue for or against classifying this case as true convergent evolution versus parallel artificial selection.
4. Art: Research the rose family's horticultural history of ornamental purple-leaf cultivars, then create a comparative color-swatch study of this tree's purple against Crimson King maple's purple, noting any visible difference in hue or saturation.
5. Evaluate: Submit the written argument and the comparative color-swatch study.

Discussion questions

What vocabulary should biologists use to distinguish human-directed trait selection from unguided natural selection, if any is needed?

Assessment

Rubric covering argument clarity and precision of the color-swatch comparison.

Extension

Compare this case to schubert cherry's naturally-occurring (not bred) purple pigment shift.

SOURCES & METHODOLOGY

Background research drawn from TreeWalk NYC's own species research. Lesson structure (objective, background, materials, procedure, assessment, extension) follows the format used by real, freely published environmental-education lessons, including the National Park Service's "Looking at Leaves" 5E lesson plan (Acadia Outdoor Classroom Collaborative), Penn State Extension's "Sorting and Classifying Leaves," and Project Learning Tree's leaf-activity guidance -- adapted and written fresh for this species, not copied from those sources. This is an educational pilot; standards references indicate topic alignment, not formal certification.